

Seminor 6

Secțiunea unei relații după o submulțime

Recapitulare =

Fie $f = (A, B, R)$ o relație binară, unde $R \subseteq A \times B$

Fie $x \subseteq A$ și $y \subseteq B$

$$f(x) \stackrel{\text{def}}{=} \{b \in B \mid \exists x \in x \text{ a.t. } xfb\}$$

$f(x) = B$

↑ secțiunea relației f după subm. x

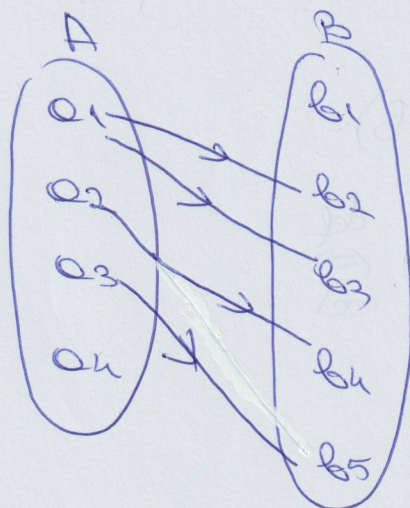
$$f^{-1}(y) \stackrel{\text{def}}{=} \{a \in A \mid \exists y \in y \text{ a.t. } afy\}$$

↑ rețracția relației f după submulțimea y

(32) Fie mult. $A = \{a_1, a_2, a_3, a_4\}$, $B = \{b_1, b_2, b_3, b_4, b_5\}$
 $x = \{a_2, a_4\}$, $y = \{b_1, b_2, b_4, b_5\}$ și rel.

$$R = \{(a_1, b_2), (a_3, b_5), (a_1, b_3), (a_2, b_4)\} \subseteq A \times B$$

Det mult. $R(x)$, $R\langle a_2 \rangle$, $R^{-1}(y)$, $R^{-1}\langle b_5 \rangle$, $R(B)$, $R^{-1}(A)$



$$R(x) \stackrel{\text{def}}{=} \{b \in B \mid \exists x \in x \text{ a.t. } xfb\} = \{b_4\}$$

$$R\langle a_2 \rangle = \{b \in B \mid \exists x \in \{a_2\} \text{ a.t. } xfb\} = \{b \in B \mid a_2fb\} = \{b_4\}$$

$$R^{-1}\langle y \rangle = \{a \in A \mid \exists y \in y \text{ a.t. } afy\} = \{a_1, a_2, a_3\}$$

$$R^{-1}(\{b_5\}) = \{a \in A \mid \exists y \in \{b_5\} \text{ s.t. } a f y\} = \{a \in A \mid a f b_5\} \\ = \{a_3\}$$

$$R^{-1}(B) = \{a \in A \mid \exists y \in B \text{ s.t. } a f y\} = \{a_1, a_2, a_3\}$$

$$R(A) = \{b \in B \mid \exists x \in A \text{ s.t. } x f b\} = \{b_2, b_3, b_4, b_5\}$$

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(34) Sei $f = \{(x, y) \in \underbrace{\mathbb{R}}_{\text{dom}} \times \underbrace{\mathbb{R}}_{\text{codom}} \mid x^2 + y^2 = 1\}$

a) Sei $x = [-2, \frac{1}{2}]$ u. $y = [-\frac{1}{2}, 1]$ def $f(x \cap y)$, $f(x) \cap f(y)$

$$f(x \cap y) = [-1, -\frac{\sqrt{3}}{2}] \cup [\frac{\sqrt{3}}{2}, 1]$$

$$f(x) \cap f(y) = [-1, 1]$$

$$\stackrel{!}{=} [-1, 1]$$

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stetig

(35) Sei $f = (A, B, R)$ u. $f' = (A, B, R')$ u. sei $x, x' \in A$
Sei a dem B :

b) $(f \cup f')(x) = f(x) \cup f'(x) \quad (\subseteq B)$

Sei b beliebig, $b \in (f \cup f')(x) \stackrel{\text{def}}{=} \text{rect}$

$\Rightarrow \exists a \in x \text{ s.t. } a f b \stackrel{\text{def}}{=} \text{"u"}$

$\Rightarrow \exists a \in x \text{ s.t. } a f' b$

$$b \in f(x) \cup f'(x) \stackrel{\text{def}}{=} b \in f(x) \vee b \in f'(x) \stackrel{\text{def}}{=}_{\text{set}}$$

$$\Leftrightarrow (\exists a \in x \text{ a.t. } (a, b)) \vee (\exists a \in x \text{ a.t. } (a, b'))$$

$$(*) \quad \boxed{\exists x (A \vee B) \Leftrightarrow \exists x A \vee \exists x B}$$

$$\Leftrightarrow \exists a \in x (a, b \vee a, b')$$

$$\exists x (A \wedge B) \Rightarrow \exists x A \wedge \exists x B$$

(36) Fie $f = (A, B, R)$ o rel. Spec UASE
(cum oform sunt ech)

$$(ii) \Delta_A \subseteq R^{-1} \circ R$$

$$(i) \forall x \in A, R(x) \neq \emptyset$$

$$(i) \Rightarrow (ii) \text{ Vom să dem că } \Delta_A \subseteq R^{-1} \circ R$$

$$\text{Fie } a \in A \text{ arbitrar} \Leftrightarrow (a, a) \in \Delta_A$$

$$\text{Vom să dem că } (a, a) \in R^{-1} \circ R \stackrel{''0''}{\Leftrightarrow}$$

↓
I

$$\Leftrightarrow \exists x \in \overset{(A)}{B} \wedge \overset{(R^{-1})}{B} \text{ a.t. } a R x \text{ și } x R^{-1} a \Leftrightarrow$$

$$\Leftrightarrow \exists x \in B \text{ a.t. } a R x \wedge a R x \stackrel{\text{idempotență}}{\Leftrightarrow}$$

$$\boxed{\Leftrightarrow \exists x \in B \text{ a.t. } a R x}$$

$$\text{În (i) luăm } x = a \Rightarrow R(a) \neq \emptyset \Rightarrow \exists b \in R(a) \subseteq \\ \Rightarrow \exists b \in B \text{ a.t. } a R b$$

!!! term 36, 37, 38, 39, 40

$$(ii) \Rightarrow (i)$$

Fix $x \in A$ arbitrary. Wenn $R_{<x>} \neq \emptyset \Leftrightarrow$

$$\Leftrightarrow \exists b \in B \text{ s.t. } b \in R_{<x>} \Leftrightarrow$$

$$\Leftrightarrow \exists b \in B \text{ s.t. } x R b$$

$$\text{Cum } x \in A \Rightarrow (x, x) \in \Delta_A \xrightarrow{(ii)} (x, x) \in R^{-1} \circ R \stackrel{\text{def}}{=} \text{"0"}$$

$$\Leftrightarrow \exists y \in B \cap B \text{ s.t. } x R y \text{ s.t. } y R^{-1} x \Leftrightarrow$$

$$\Leftrightarrow \exists y \in B \text{ s.t. } x R y \text{ s.t. } x R y \Leftrightarrow$$

$$\boxed{\Leftrightarrow \exists y \in B \text{ s.t. } x R y} \quad \square$$

(ii) $\forall A'$ mult., $\forall P_1, P_2 \subseteq A' \times A$, dann $(R \circ P_1) \cap (R \circ P_2) \neq \emptyset$ atunci $P_1 \cap P_2 \neq \emptyset$

$$(ii) \Rightarrow (i) \quad (i) \Rightarrow (ii)$$

Fix $x \in A$ arbitrary $R_{<x>} \neq \emptyset \Leftrightarrow \dots \Leftrightarrow \exists b \in B \text{ s.t. } x R b$

In (ii) luăm $A' =$

$$P_1 =$$

$$P_2 =$$

$$\subseteq$$

$$\subseteq$$

Conform (ii) $\Rightarrow \forall A'$ mult., $\forall P_1, P_2 \subseteq A' \times A$ dacă $P_1 \cap P_2 \neq \emptyset$ atunci $(R \circ P_1) \cap (R \circ P_2) \neq \emptyset$